Kyle Aaron Thompson

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Education

Auburn University — Auburn, Alabama
Ph.D. Mechanical Engineering — *Expected 2028*
M.S. Mechanical Engineering — *August 2025*
B.S. Mechanical Engineering, Magna Cum Laude — *August 2020*

Experience

Graduate Research Assistant – Auburn University — August 2020 – Current
GPS and Vehicle Dynamics Laboratory (GAVLAB) — *Auburn, Alabama*

- Conduct sponsored research in guidance, navigation, and control (GNC) of autonomous ground vehicles, focusing on guidance and control, with additional work on aerial platforms.
- Serve as controls, guidance, and coding lead for Autonomous Tiger Racing (ATR), Auburn's Indy Autonomous Challenge (IAC) team.
- Built the team's raceline optimizers and optimal online feedback controllers, and developed the majority of the current software stack.
- Developed a GPS-independent leader–follower convoy system using UWB ranging and vehicle models, achieving 1–5 m path-duplication accuracy at 10–50 m following distances.
- Designed an intelligent emergency stop system that tracks minimum-jerk trajectories with an LQR path-tracking controller, targeted to run on a low-cost backup processor.
- Published multiple academic papers as first author and co-authored additional conference and journal papers.
- Assisted in the laboratory instruction of undergraduate courses taught at Auburn University.

United States Marine Corps Reserves — July 2014 – Current
Gunnery Sergeant

- Responsible for safe operation of Marine Corps artillery weapon systems at Battery K, 2nd Battalion, 14th Marines.
- Responsible for the training and supervision of an Operational Firing Platoon (approximately 35 Marines).
- Built and instructed curriculum used in training programs to grant professional certifications for Rocket Artillery Operators.
- Deployed to Afghanistan with Task Force Southwest 19.1 and 19.2 (2019) and to Okinawa (2022–2023).

Teledyne Brown Engineering, Inc. — April 2015 – July 2015; June 2016 – August 2016
Classified Lab Administrative Assistant — *Huntsville, Alabama*

- Responsible for opening and securing the classified lab after-hours and on weekends, logging of lab and maintenance activities, and general enforcement of government and company security policies.
- Supported generation of software documentation.
- Responsible for performing basic system administration functions in support of users in the classified lab.

Selected Publications

- **K. Thompson**, "Intelligent Emergency Stop System for Autonomous Ground Vehicles Using Minimum Jerk Trajectories and Path Tracking Linear Quadratic Regulator," *M.S. thesis*, Auburn University, Auburn, AL, Aug. 2025.
- **K. Thompson**, B. Jones, S. Martin, and D. Bevly, "GPS-Independent Autonomous Vehicle Convoying with UWB Ranging and Vehicle Models," *NDIA Ground Vehicle Systems Engineering and Technology Symposium (GVSETS)*, Novi, MI, Aug. 2022.

Skills

Computer and Software: Windows (10, 11), Linux, macOS, LaTeX, Visual Studio, Xcode, Single Board Computers and Microcontrollers
Programming and Tools: C++, MATLAB, Python, C#, Kotlin, Swift, Qt, ROS, and SolidWorks (3D CAD)

Achievements and Involvements

- Dean's List – Fall 2015, Fall 2016, Spring 2017, Fall 2018, Spring 2020 (Auburn University)
- Eagle Scout – Boy Scouts of America, 2013 (Troop 8 – Madison, Alabama)